Rescue Operations System (ARDMS) - Deployment Guide

This document exclusively covers the setup, network configuration, and compilation protocols for the ARDMS emergency management network.

1. Prerequisites

Ensure the host deployment machine (Local Server) has the following installed:

- **Node.js**: Version 18+ (for the backend server)
- **Python**: Version 3.8+ (for synchronization and build automation scripts)
- **Java Development Kit (JDK)**: Version 17 (Required by React Native/Gradle)
- **Android SDK**: Available via Android Studio (with environment variables `ANDROID_HOME` correctly set)

2. Setting Up the Local Backend

1. **Initialize Dependencies**:
   Navigate to the `system-backend` directory and install NPM packages:

   ```
   cd system-backend
   npm install
   ```

2. **Launch the Server**:

   ```
   node server.js
   ```

   The backend will auto-detect the local Wi-Fi/LAN IP address (e.g., `192.168.x.x`) and bind to it. It will also serve the static Web Admin dashboard at `http://<IP_ADDRESS>:3000`.

3. Synchronizing Frontend Edits

The React Native mobile apps directly wrap HTML files located in the root directory. If you modify `preview-mobile-app.html`, `preview-rescuer.html`, or `raw_admin.html`, you MUST synchronize these files into the React Native build pipeline before compiling.

Run the synchronization script from the root project directory:

```
python sync_apps.py
```

This script will parse the HTML, inject the detected Local IP Address into the JavaScript payloads, and copy them into the respective React Native project folders as `htmlStr.js`.

4. Compiling the APKs

After synchronization, you can compile the APKs using the provided automated Python scripts. These scripts orchestrate Gradle and React Native CLI commands.

- **Build Public App**:

  ```
  python build_public_apk.py
  ```

- **Build Rescuer App**:

  ```
  python build_rescuer.py
  ```

- **Build Admin App**:

  ```
  python build_admin_apk.py
  ```

(Note: Compiling the Admin APK is only necessary if you require tablet deployments for commanders. Otherwise, they can just use the Web Admin via a browser).

Upon successful completion, the compiled, release-ready APKs will be automatically moved to the `Output_APKs/` directory.

5. Network Reconnection & App Configurations

All compiled APKs feature an offline-capable manual IP entry system.

- If the central backend server changes its Wi-Fi network (and therefore its IP address), **you do not need to recompile the APKs**.
- Operators and citizens can simply tap the server connection status icon in the app, type the new backend IP address, and seamlessly reconnect to the websocket pipeline.